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ECEN 5224

Week 11 Lab Report

04/01/26

*Lab 1*

**Objective:**

The objective of this lab was to use HyperLynx to run Eye Diagram simulations of differential drivers and a differential pair transmission line. The stackup is modified to use a value of Dk = 4 for the inner dielectric layers. With each simulation, the height and width of the eye is recorded and measurements are compared to various schematic layouts.

**Method:**

HyperLynx was used to simulate a transmission line circuit with properties: Dk = 4 for the inner dielectric layers, Dk = 3.3 for the TOP and solder mask layers, a ~100Ω differential impedance, and with a T-Termination. Two circuits were compared, one with matched trace lengths and the other with a 0.2 ns delay, both using the CMOS, 3.3V, ULTRA, OUT driver for the P and N sides. The eye diagrams were plotted, measured, and analyzed.

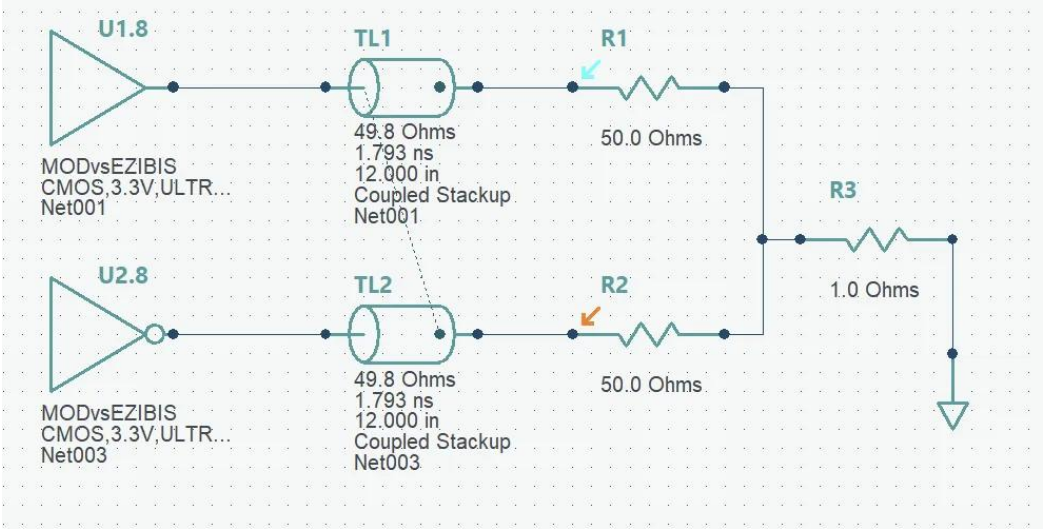



*Figure 1: Matched Differential pair with no length mismatch*

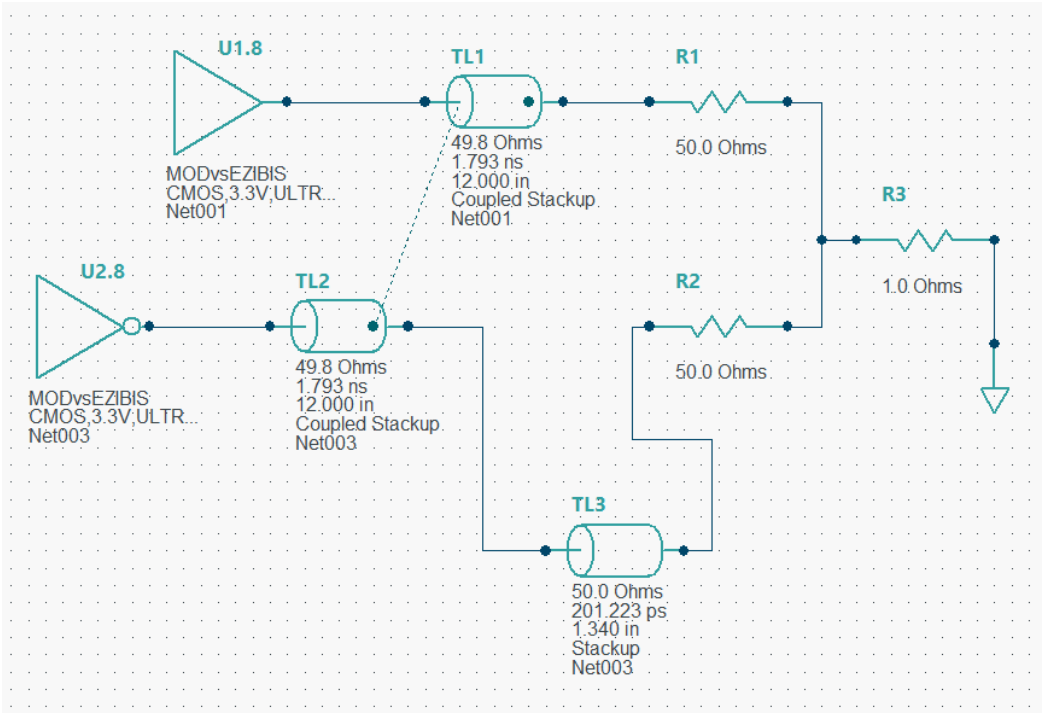



*Figure 2: Additional 50 Ω, 201.2 ps line (TL3), on the negative leg to introduce a 0.2 ns length mismatch*

**Results:**

Eye diagrams were captured for the differential signal at the load. Measurements were taken using the cursor tool at the eye's crossing points.

| Parameter | No Mismatch | 0.2 ns Mismatch |
|---|---|---|
| Eye Height (V) | ~6.00 | ~6.00 |
| Eye Width (ps) | 987.5 | 964.2 |

*Table 1. Measured eye diagram parameters*

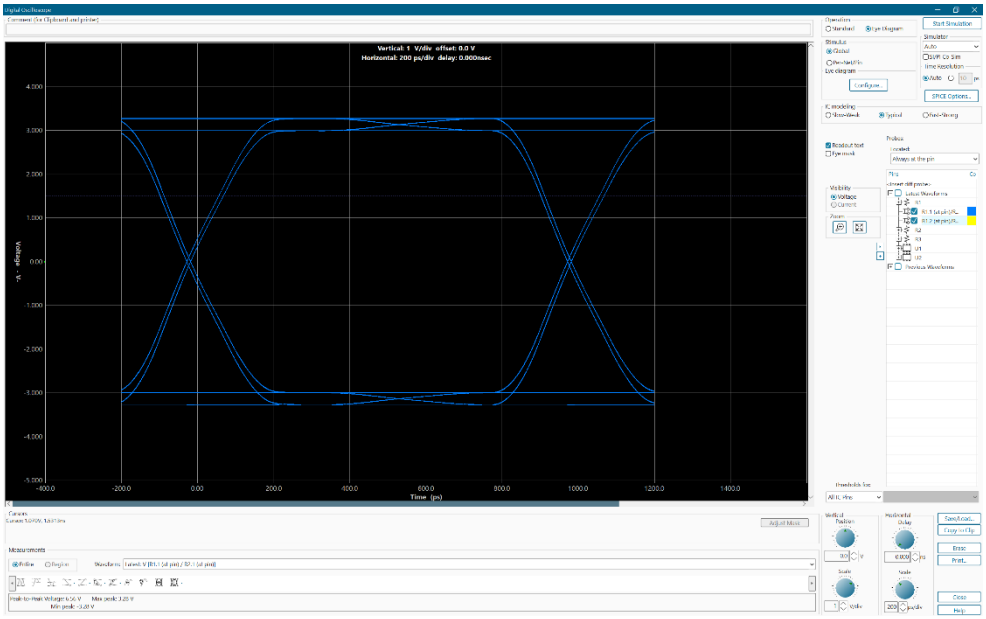


*Figure 3. Eye diagram with no length mismatch. Clean crossing, height ~6 V, width ~987.5 ps*

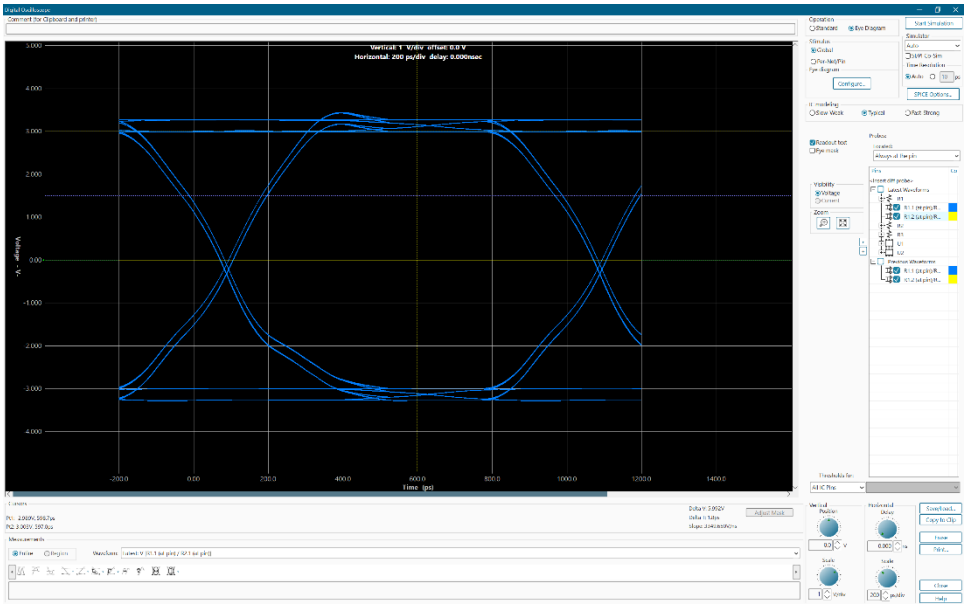


*Figure 4: Eye diagram with 0.2 ns mismatch. Slight skew, height ~6 V; width ~964.2 ps*

**Analysis and Conclusion:**

The first circuit with matched trace lengths has an eye diagram that shows clean, symmetric crossings with a large width. Introducing the 0.2 ns delay on TL3 causes a small timing skew between the positive and negative lines. This degrades the differential signal quality. The eye height remains essentially unchanged at ~6 V, however, the eye width narrows by approximately 23.3 ps. This reveals the added jitter from the skewed arrival times of the two signals. The added delay of ~201 ps (1.340 in of 50 Ω trace) skews the crossings of the negative leg in time, causing the eye to open later and close sooner. This is consistent with what is expected to happen. A length mismatch in differential pairs convert common-mode noise into differential noise and reduce timing margin.

This lab shows that even a routing mismatch under 1.4 inches is enough to introduce noticeable jitter at 1 Gbps. As data rates increase, this 23 ps change becomes a larger fraction of the bit period. This means that a mismatch that appears acceptable at low speed can become a hard failure at higher rates. This is why high-speed interfaces like PCIe enforce tight differential pair length matching rules, usually within mils.

<u>*Lab 2*</u>

**Objective:**

The objective of this lab is to learn two aspects of differential signaling. The first being the effect of different termination schemes on eye quality (Cases 1 to 3) and the second being the effect of increasing data rate on a pair with a fixed 0.2 ns length mismatch.

**Method:**

Three termination configurations were simulated using coupled ~50 Ω transmission lines with a length of 12 inches. The stackup remains the same as lab 1. The three cases varied the termination at the load end:

- Case 1: Single 100 Ω differential termination (R1 across the two lines)
- Case 2: Individual 50 Ω terminations on each leg to ground
- Case 3: Open-ended (no termination)

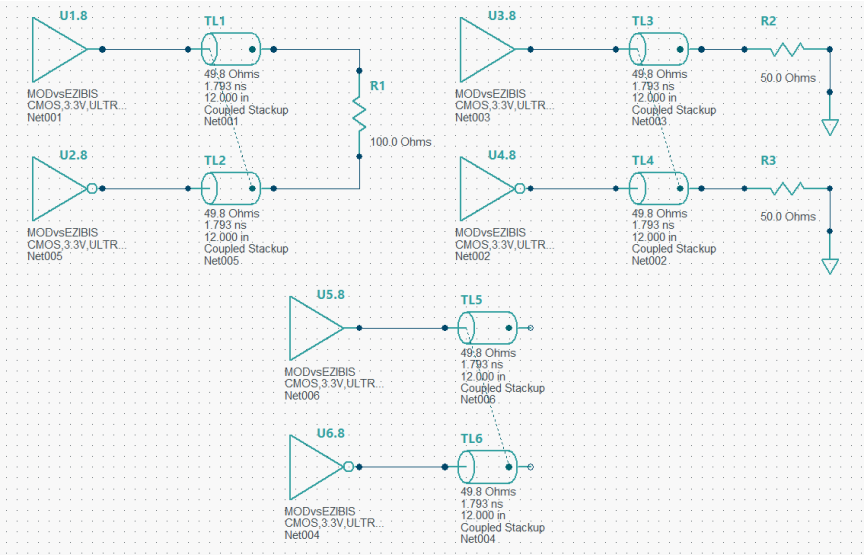



*Figure 5: Circuit schematic for Cases 1-3. Top left pair (TL1/TL2) -> 100 Ω differential (Case 1). Top right pair (TL3/TL4) -> individual 50 Ω to GND (Case 2). Bottom pair (TL5/TL6) -> unterminated (Case 3)*

A separate circuit with a 0.2 ns length mismatch (same topology as Lab 1 Circuit 2, but with a 100 Ω differential termination) was swept across data rates of 1 Gbps, 1.75 Gbps, 2 Gbps, and 2.25 Gbps using a PRBS-7 stimulus to observe eye closure.

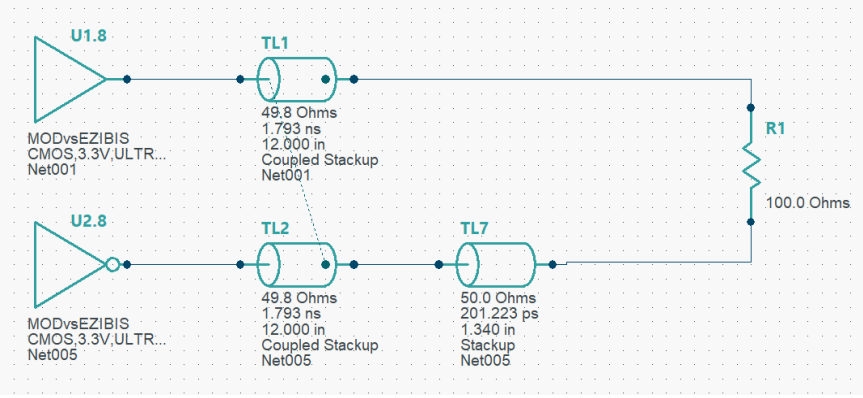



*Figure 6. Circuit for data rate simulations. TL2 path includes TL7 (50 Ω, 201.2 ps) for 0.2 ns mismatch, 100 Ω differential termination (R1)*

**Results and Plots:**

| Case | Termination | Eye Height (V) | Eye Width (ps) |
|---|---|---|---|
| 1 | 100 Ω differential | 6.109 | 990.2 |
| 2 | 50 Ω each leg to GND | 5.969 | 988.4 |
| 3 | Unterminated | Undefined | Undefined |

*Table 2: Lab 2 measured eye parameters for Cases 1-3*

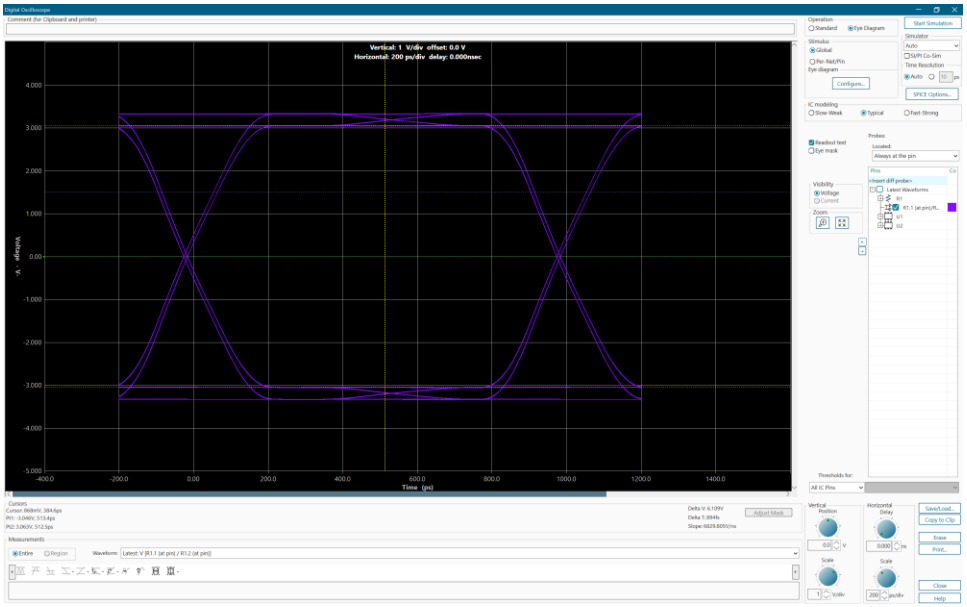



*Figure 7: Case 1 eye diagram (100 Ω differential termination). Height: ~6.109 V, Width: ~990.2 ps*

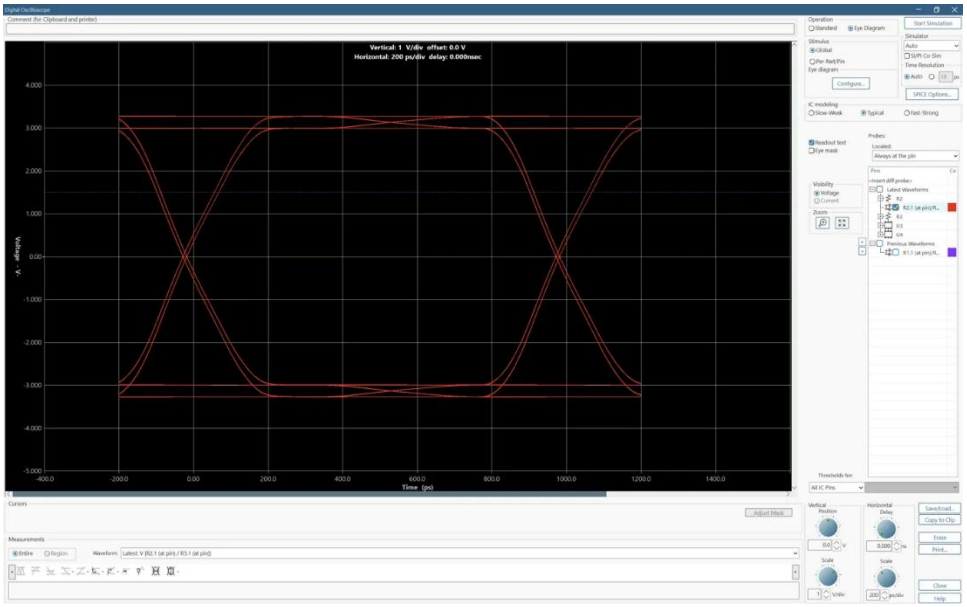



*Figure 8: Case 2 eye diagram (50 Ω each leg). Height: ~5.969 V, Width: ~988.4 ps*

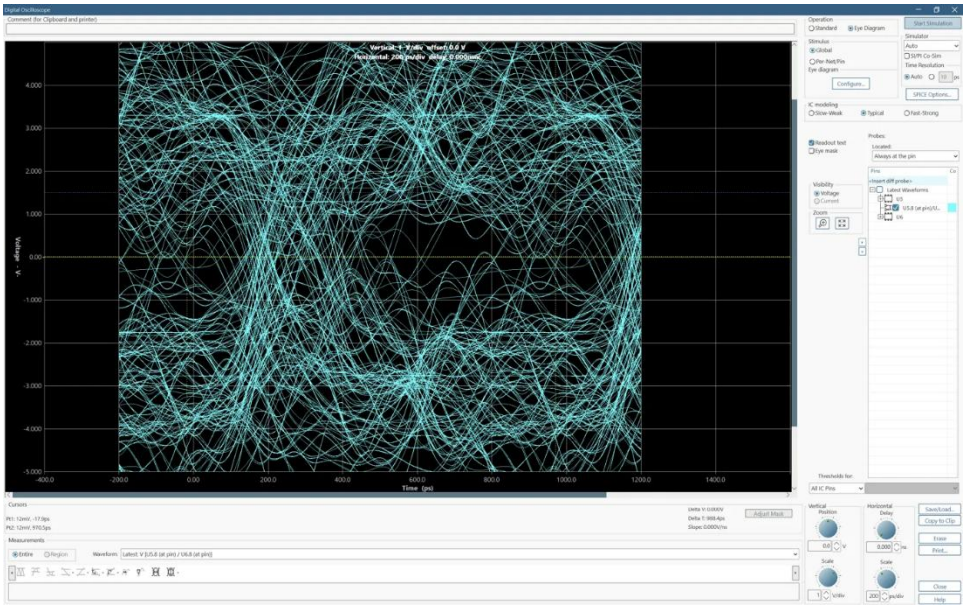


*Figure 9: Case 3 eye diagram (unterminated). Height: Undefined V, Width: Undefined*

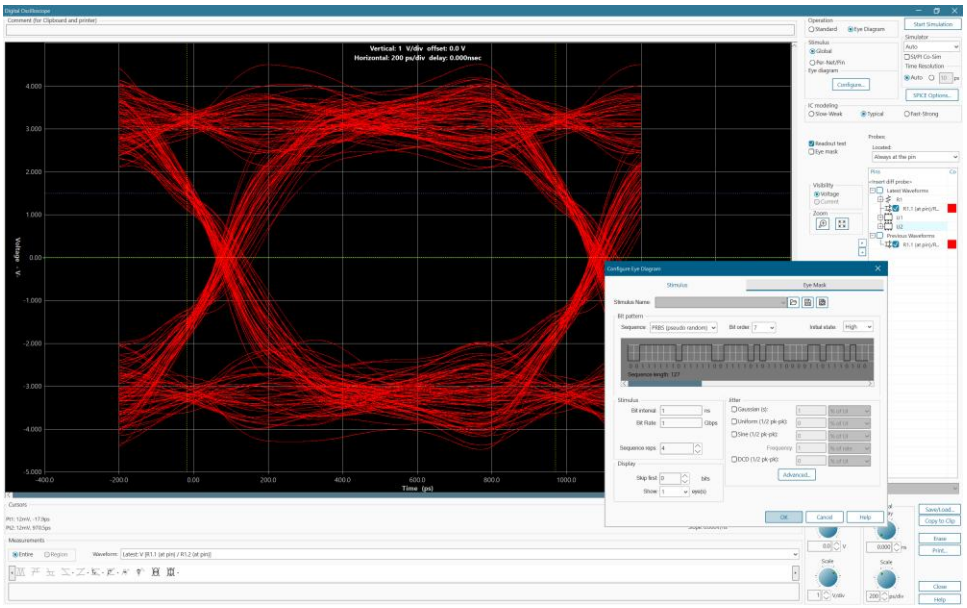


*Figure 10: Eye diagram at 1 Gbps. Open eye. Moderate jitter due to length mismatch*

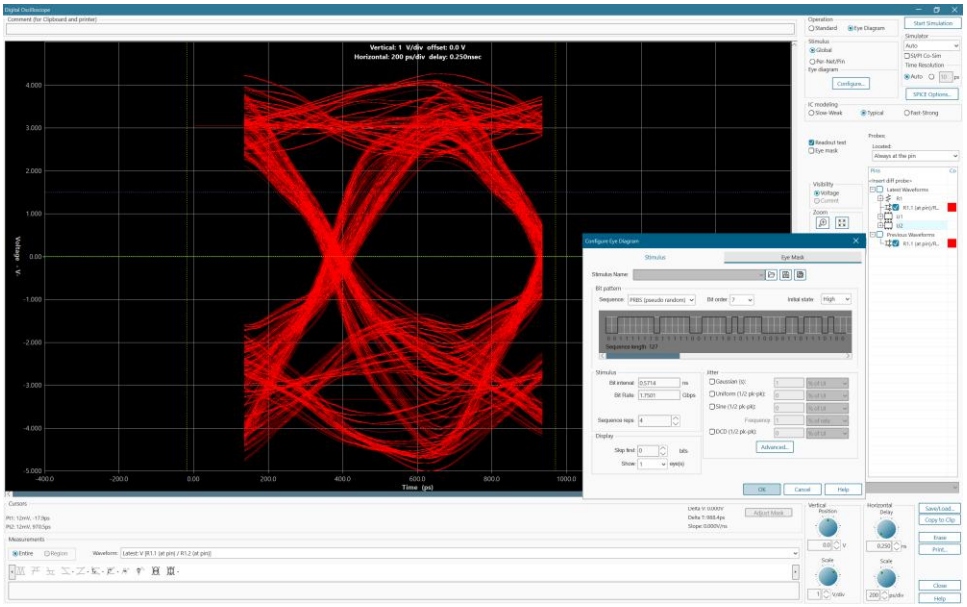


*Figure 11: Eye diagram at 1.75 Gbps. Eye begins to close. More crossing noise*

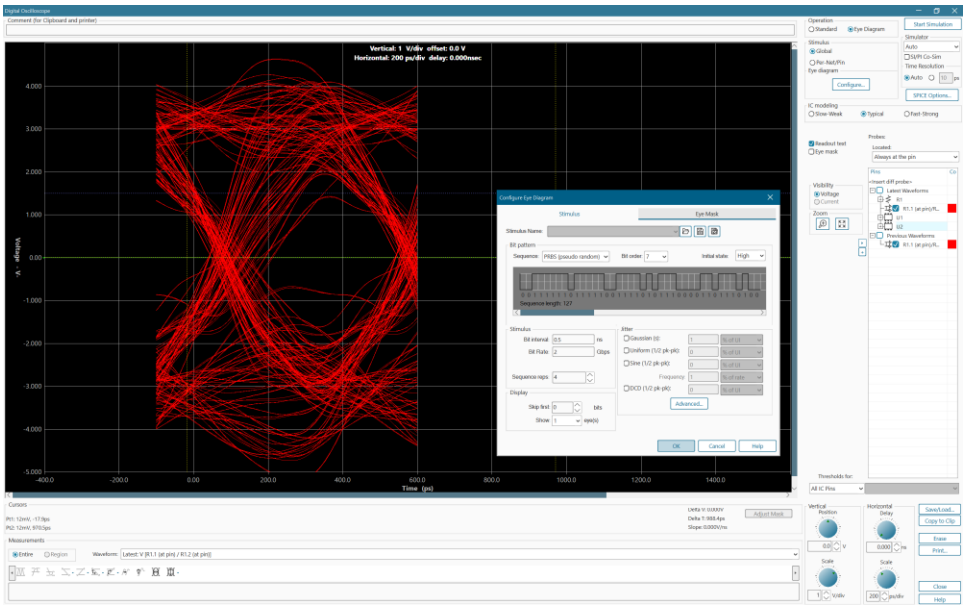



*Figure 12: Eye diagram at 2 Gbps. Significant eye closure. Mismatch represents a larger fraction of the bit period*

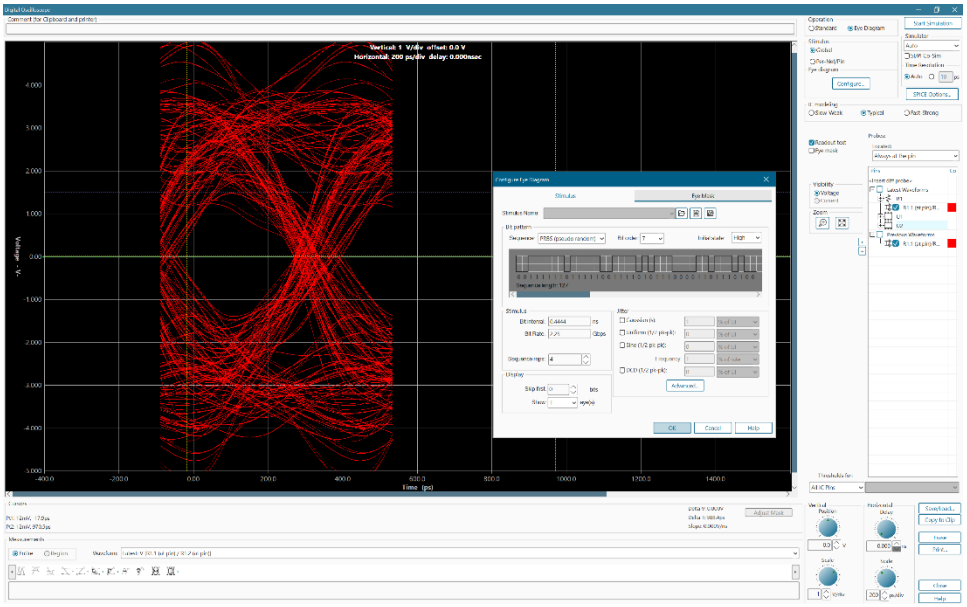



*Figure 13: Eye diagram at 2.25 Gbps. Eye essentially closed. Data would be unreliable*

**Analysis and Conclusion:**

All three cases produced similar eye heights of ~6 V, but the different termination schemes affects the reflections and signal integrity.

Case 1 (100 Ω differential) is the most common approach for differential lines and provides clean matching with no net current drawn in the steady state. It produced a height of 6.109 V and a width of 990.2 ps.

Case 2 (individual 50 Ω to ground) effectively presents 100 Ω differentially but also provides a DC return path for each leg. This introduces a slightly lower eye height (5.969 V) and marginally reduced eye width (988.4 ps). Most likely due to asymmetric reflections from the ground-referenced terminations.

Case 3 (unterminated) produced the same eye parameters as Case 1 (6.109 V, 990.2 ps), which appears counterintuitive at first. At the bit rate used (1 Gbps), the round-trip reflection from the open end may have arrived after the measurement window, or the driver output impedance alone was sufficient to suppress reflections. A higher data rate or longer trace would be expected to show more severe degradation with an open termination.

For the fixed 0.2 ns length mismatch, increasing the data rate progressively closes the eye. At 1 Gbps (1 ns bit period), the 0.2 ns skew represents 20% of the bit period. So already notable but the eye still remains open. At 1.75 Gbps (~571 ps period), the skew is ~35% of the bit period, and significant eye closure is visible. By 2 Gbps (500 ps period), the skew is 40% of the bit period and the eye is substantially degraded. At 2.25 Gbps (~444 ps), the eye is essentially closed and reliable data detection would be hard to achieve. This demonstrates the key practical constraint of this lab. A length mismatch that is acceptable at one data rate becomes a hard limit at higher speeds. The ratio of skew to bit period determines the eye distortion. This means that routing constraints should be designed with the target data rate in mind from the very beginning.

These simulations demonstrated that differential pair routing and termination both play a role in signal quality. A 100 Ω differential termination is the preferred approach, revealing the cleanest eye. More critically, the data rate sweep showed that the same 0.2 ns length mismatch that leaves the eye open at 1 Gbps completely closes it by 2.25 Gbps. This highlights that mismatch is not a fixed penalty. It is something that scales with the data rate, making differential pair length matching a necessary design process for any high-speed interface.